

Active Inference Institute ~ 2024 Quarterly Roundtable #4

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Hello, welcome. It's December 19th, 2024. We're in our fourth quarterly roundtable for the year. So, welcome to the Active Inference Institute. We're a participatory online institute that is Communicating, Learning, and Practicing Applied Active Inference. This is recorded in an archive live stream, so please provide feedback and super chats, live chats, so we can improve our work. All backgrounds and perspectives are welcome and we'll be following video etiquette for live stream. Everything at the institute is participatory, so head over to activeinference.institute to get involved and check out our updated site. This slide has a lot of stable links, which you can go to as an entry point, and we'll always keep these stable links updated even as their underlying targets evolve, so that you can have some stable reference and entry points into it. Okay, this is our fourth quarterly roundtable of the year. It is close to the end of the year. We've done a quarterly roundtable every quarter since the beginning of 2021. So we're approaching 16, a tet of tets. Alex and Blue, co-officers of the year, would you like to say hello and anything else?

Hi, happy 2024, happy end of 2024. I'm excited about all the things that the institute has done this year and about all the changes that are happening and transitions that we're looking forward next year. Hi, and this reminds me how it started. Now we are institute, but I remember how it was a comms group in just online format without even possibly knowing names or faces. So we are in the beginning of the year. But from the beginning, understand what active inference approach and principles could help us to grow into something great.

Thank you. Yes, Teamcom 2020, active lab 21, institute 22. All right, let's continue on first at the institute scale.

All right, one of the major highlights of this year being the second version snapshot release of the sort of keystone publication, active inference ecosystem and institute one way or the other. And the first version of this publication, we did also in a great collaborative team in 2023.

This year, we dipped back in, we started with where we left off, as any updater would, and took it in a ton of different directions.

Ecosystem.active inference institute is the entry point. And just to give a bit of the anatomy of the document and show a few of the pages, some of the nested memes for nested themes.

Ecosystems. At the top level, this living documents or production has an abstract, a list of authors, a single page on active inference, a closing discussion, and two main body sections, one on the institute and one on the ecosystem.

About the institute, the subsections include history, mission, vision, values and principles, our focus areas, some of our directions, strategy,

the organization of the institute, communications, and other topics.

About the organization.

Under that, if you unfold that tick mark, you'll see the formal morphology.

For example, the scientific advisory board, board of directors, officers, members, the organizational units, administrative functions, the educative education functions, the re-inference research functions, and then a variety of programs that we have at the institute, such as volunteer, internship, mentorship, so on.

And some having begun in the beginning, others being incipient.

In the institute project section, there's a roster of projects, different types, different stages, and other projects, different types of projects, different types of projects, different types of projects, and different types of projects.

So, what is happening? What has happened?

These are some of the things that have happened right very here.

On the ecosystem side,

the sections include the priorities and the challenge areas for the broader space, the fields, where people are really out there, applying active inference and adjacent techniques.

We're starting to explore how we think about the development of the ecosystem, about how to curate and stay on top of the different projects that are out there.

Past, present, and future, likely overwhelmingly not at the institute.

However, we see a lot of value and meaning in having domain experts and domain curious contributors who contribute in a really nuanced way, for example, to a certain field or subfield, having something like an ongoing research report or meta-analysis of different approaches, for example, in robotics.

And those sorts of sector reports can be great links to share for onboarding, for starting with somebody who wants to enter into active inference and their field of reference.

Also, it could support research, students, people who are applying or thinking about entrepreneurship or startups

in different areas.

So those are some of the pages.

And one of the big jumps this year compared to last year, last year we wrote it linearly in a Google Doc.

This year we went full nested collaborative style.

And while it takes some time to get your fractal legs running, it's going to enable a lot of very expressive ways for us to collaborate and compose and translate and generate documents.

So overall, it's awesome to have a team, have a lot of eyes on different parts of the documents, and for the process and the product to be made for people to use it.

So we hope that you enjoy diving around that document.

Alex or Blue, any comments on that?

Alex or Blue, any comments on that?

Yeah, I think this is a really neat transition from the linear Google Doc to the interactive coda. And it reminds me of a wiki, right? Like a working wiki, a living, breathing wiki document.

When we were at SFI, the project wiki that everyone had and everyone could contribute to, I think that putting the institute projects there is really nice and helpful so you can kind of see what's going on and when was this page last updated and all of those things. So I think that as a living document, this is a good place for it, a nice home.

And even though it's only our second versioned snapshot, first off, now we can make many more micro edits along the year and then making the snapshotting a lot more simple. And it's already become something I look forward to in terms of our nested timescales and the annual and slower. Maybe we'll have a decade scale snapshots.

Okay, another major highlight from the end of this year on November 13th, 14th, and 15th was the fourth Applied Active Inference Symposium. This was quite an event. There were three days of activities. The first two days had two mega streams. The third day with John Klippinger and the BioFirm project, leading a workshop on BioRegional Active Inference and AI. There was a team of co-organizers. So it was great to work with people on different fronts, including people who were co-organizers as well as presenters. There was 30 plus hours of people sharing what was happening this year in Applied Active Inference, which is

coming truer and truer to what this symposium has always been about. Not that application is the only thing out there, but people are always asking and wanting to know how and where and why and what about doing Active Inference. And this brought together a lot of presenters from around the world who are sharing where they're at on those journeys and the tools and the approaches and everything along the way. So I look forward to the symposium for 2025. It's never too early to begin. If someone's interested in joining, we can begin with a template and some of the notes that we used this year. We can work with a local partner possibly to explore some kind of in-person activities regionally. We'd be very interested in some sponsors or other partners who might like to support the open science publishing elements or any other number of things. So it's not too early if you're interested in the symposium or you thought it was cool or you thought there was something that could be updated or improved or that its reach or impact could be magnified. Let's do that next year. Any comments on the symposium? All right. Now into some updates about different programs and different personnel.

About the Research Fellows Program. Research Fellows are individuals affiliated with the Institute reflecting the alignment of their research and our open science approach. This was a new program we began this year. So it still counts to include it in our year-end review. And also in the fourth quarter we added two new research fellows, David Blumen and Robert Warden. We will be in contact with all the fellows hearing about how we can support them best for the coming year. One exciting avenue is that the donate link supports the project restricted donations and so fellows we hope will be in a position to receive donations to their project and then have the ability to decide how those are used in service of the project with the oversight and streamlined overhead from the Institute side. So we hope that we can provide that kind of formal financial organizational support to research fellows possibly eventually much more than that and also work more closely next year with the scientific advisory board and with interns to to really accelerate their projects. The next round of review of fellows will be sometime in 2025. We will

while we look to continue to learn organizational formal procedures and work with the board of

directors to have that level of formal oversight and governance. In the beginning of next year, we'll be electing the next cohort of the board of directors. So do consider applying, especially if you have experience in any of these areas like nonprofits or below.

About partnerships, partnerships are structured relationships between the institute and other organizations who are aligned and are looking to make meaningful, visible, impactful contributions to the active inference open source ecosystem, who are looking to upskill and accelerate their own processes of learning and application. Organizations who are looking to connect with our community and the growing field. Possibly partners who are looking to form a regional or language or sector or domain level collaboration and also provide critical input to the institute and question mark and beyond. The second half of this year was marked by the addition of, well first, this program arising in 2024, which is cool. And a second partner joining, Newman Games, thinking a lot about active inference in and for games. And also some discord fun. With our partner first principles first, we have coordinated on aspects of organizing the symposium and some other activities as well. So there's a lot of ways that we look to develop this out.

So if you're applying active inference in a company or you're thinking about doing so or not, this could be a very interesting opportunity. And we will, as with all things, continue to update it based on how our niche responds and reflects back these affordances. Alex or blue?

Okay. Also in big 2024 news was the recognition of our 501c3 tax exempt status as a nonprofit in the US. So we aim to develop our philanthropy program and endowment over the years.

As we develop from our current all volunteer status on through something that's larger and more sustainable for people's time and reflects the relevance of what they're doing.

So we're going to be doing.

Please head over to donate.activeinference.institute if you feel like making such a donation or get in touch. If you have any other custom considerations. If you'd like a conversation about where that could go, we would look forward to it as we deeply respect the priceless ways that people attend to the work and think it could go much, much, much bigger and faster. Not that size and speed are everything, but with financial support. Alex or blue? Do you have any comments?

That was beautifully put.

Truly, I do not like to dilute our act inf semantics with financialization. So I hope that we can find the right kinds of support through grant and philanthropy and so on that allow us to just pitch that fastball and call those shots and have the opportunity for people to provide crowdsourced support of many kinds, but really hold up the priceless meaning of attention while also keeping it compliant and sustainable and reasonable organizationally. And again, as alluded to in the fellows page, you can go to that link and drop down to make a project restricted donation, which we can now account for. So those donations will have a reasonable institute level overhead percentage taken. The rest will be available to the project for whatever they feel like is going to facilitate the work and it's a cool way to support specific projects or let us know if you want to support a specific project that's not here right now.

Okay.

Okay, some quick updates from the organizational units. First on Eduactive. So we had four main projects this year, all of which will submit in due time their measurements and re-prepare for next year. Active inference ontology, audio visual production, textbook group activities, and the active inference journal. Just briefly on the textbook group, this is working through and sharing an epistemic niche around the 2022 active inference textbook. Still the only textbook out there, but likely not for long. Cohort six finished earlier this year. Cohort seven will continue from the midpoint on in 2025. And as with all things, this is a volunteer activity.

So if somebody wants to see a time zone, an open discussion, application oriented sessions, if somebody wants to scribe or facilitate in live sessions, if somebody wants to make asynchronous contributions, like augmenting or cleaning up questions, the hundreds of questions that are written in the document, if somebody wants to explore translating parts of this open source textbook or our open source materials around it,

there's a ton of ways to get involved. So like, if you're listening, just jump in and let's do it.

Because there's a lot of people reading the book and there'll be more learners in the time to come.

So there's never been a better time to increase the opportunities for others in terms of synchronous and asynchronous active education.

And also, we will continue to

template other reading groups and other educational material, because as the years pass by, now it'll be almost three years since this textbook coming out in

March 22,

that has opened up so many other avenues for learning. So we will continue to

hold the space for this 2022 textbook

while also seeing it as kind of like an attractor or a nexus where a lot of other resources get brought in.

A lot of other things can happen there too.

So, just on the textbook group, I think that the math learning group this year that was

done by Octopus was such a big, like, augmentation for the textbook group. So shout out to Octopus for facilitating that for us. That was super helpful. And just, I want to comment on the amount of annotation, like, just, I remember like the first cohort of the textbook group, like being, you know, before the formulas were there before, I mean, you know, being among the first to do any kind of annotation for the textbook, versus the amount of annotation that's there now is like, seriously, seriously impressive. And it's amazing and awesome to see like the result of so much collaborative co-learning in like a tangible form.

So the CodaPage is really quite awesome and deep and has evolved a lot over these years.

How many people currently for cohort 7, including cohort 7, was involved at least a bit into learning active inference?

That's a great question. The number of registrations over, summed over all the cohorts is in the low hundreds,

like probably between 3 and 500. And the number of people joining one to many meetings is probably in the

few to several dozens. That...

That's a lot.

It's dozens of people who have come through, all the videos are available, um, of the recordings and, and the subsets of those people and also people who probably never joined a synchronous activity, maybe even people who didn't register and still made use of the resources, the, the YouTube views and comments and comments and discussions and all this other work. So it's just so interesting with this sort of

open science and reading group where it's like both being in the spotlight in terms of it being recorded and disseminated,

as well as not knowing what's out there. Nevertheless, though.

In audio visual production, where we coordinate the live streams and podcasts, we had a great year.

Tomorrow, Dean and I are going to do the 2024 review stream. So we'll go over, at least in a cursory quick gloss, the 137 videos that were happening this year, 2024.

Um, to, to recap some of the, the hits. These are the most popular uploaded videos on our page.

Commonality being Darius's insight series being immensely popular with the exception of other videos with Carl Fristen and the course overview for the physics of information processing with Chris Fields.

Insights took, took the leaderboard. So epic work, Darius. This was an amazing series.

For the most popular live streams,

Chris Fields course, lecture one, two, three, four, five, and six in that order. Interestingly, so it's like there's the attrition rate right there, as well as the model stream with Ryan Smith, live stream with Jonathan Gorard, which is presaging some of the connections between the Wolfram area and active inference and, and other cool work like the symposium. So for people who are skilled or not yet, but might like to be with any number of ways to help like suggesting and curating and inviting guests, facilitating streams and videos in English and other languages, improving social media posting and other ways to communicate before, during, and after videos. These are all places where somebody doesn't need to know how to derive variational free energy to make truly unique impact on the availability of these materials.

All right. To re-inference. So from the Institute side, we had several projects in re-inference.

The most active was our rxinfer.jl learning and development group around the Julia package for statistics and active inference. We continued to explore a few avenues around knowledge engineering and on the software development side in active blockference and active inference had a lot of fun.

Just about the rxinfer project.

We had many people join those enjoyable discord times, including this morning, but Thursdays this year, and the meetings, notes, tell tales. We applied for two grants as a team to support and catalyze the work that a bunch of people are involved in. We have one grant still pending. It's published though as a preprint on Farmworks, Decentralized AI Agents for Personalized Solutions, working with Vladimir Baulin and

collaborators.

So this is going to be an uncertainty to be resolved in early 2025. Something that ties in well with a lot of other projects though. So we look to continue this. And just a few days ago, we learned that we did not receive a AI safety grants village validating inference for large scale agent governance ecosystems. So if you're a grant officer or philanthropist, you're interested in AI safety, if you want to contribute, if you want to support, we have a lot of people interested in those topics. We and others believe that active inference has a unique role to play in the AI safety and alignment space. So maybe get in touch.

A little bit more generally about projects at the Institute. I'm going to take two passes at the prepare form and measure form. Here in the center, we have a holographic screen, Markov blanket, boundary, interface,

all that. And there's a cycle of preparation and measurements interfacing with that screen by Alice and Bob on both sides. And in the context of projects at the Institute, that's reflected by a prepare form and a measure

form. The measure form has a short version that basically just says, what was it? What happens?

Can we share it? And there's a longer version. So now zooming out a little bit. Preparation lets us know that you're about to start something. It might be something that doesn't ask anything of the Institute.

It's just like, Hey, list this publicly or don't, but I'm about to go out. Or maybe there's something small or even something very large that the Institute could do to support what you're proposing to do.

Or there's something that's blocking your fulfillment of that, but you'd like to propose it pending possibly a future moment when we have an internal grants program.

So that applies to every measurement from the eye saccade on up to the multi-center collaboration.

And on the measurement side, it is about what happened. It could be, I saw a bird. It could be, I saw two birds. It could be, I saw this paper. I didn't write it, but I came across this paper.

It could be, here's what we did in our project. And here's what we learned. And these preparation and measurement forms are some of the primary interfaces amongst everyone, all those listening and attending and the Institute. It's a standard that we hold Institute internal projects to as well, that they engage in this alternating two-stroke engine of reporting back, even if it's just one sentence.

But it's truly amazing how much challenge and friction and inertia there can be around reporting, even when people have done truly incredible things. So gently and kindly to myself and others,

I encourage that people fill out that measurement form when they've done stuff,

so that we can share it. If you say yes, in the newsletter and through other channels, and so that

it can be broader understood what you're doing so that you can demonstrate when you did certain things.

And it's an important channel of communication with the Institute. There was a bunch of projects that fulfilled the proposed and some measurements this year. They're all available at [activities.activeinference.institute.institute](#). And all of those projects will go inactive at the end of this year.

They will be required to fill out a measurement form and a proposal to come back online in 24.

That will support the flow of energy and attention and activity to live projects so that we can have healthy portfolio scale updating and continue to re-center the Institute as a place where people can

find out what's going on, find out what has been, find out what could be, get visibility for their work, meet others. So again, measure for providing any updates or observations.

It could just be one-off. It could be anonymous. Also for projects, it's required for internships and so on.

And prepare. So if you're proposing a new endeavor or a new project, whether you would like it listed at our activities page for visibility or whether you're seeking some further kind of support and ask big, we would be honored and excited to curate those requests and dream of a future day when we could share that with the SAB or with funders or with other groups and say, here's what people are proposing. Here's the kind of support that we would want to provide to support these people to do this work this way. And here's how we'll have accountability and follow up with the measurements too.

So just some thoughts, of course, open for everyone's co-creating and feedback and joining from there. But those are just some of the ways that we've seen how this prepare and measure two-stroke engine can help us do more together.

Alex or Blue, any thoughts on Project X or just anything else? Also, 2024 being the year that we introduced this two-stroke cycle of prepare and measure following from the 2023 course. But do you have any thoughts on that?

Yeah, I like it. The prepare and measure way of just reporting. I think it's really hard for a lot of people to toot their own horn, you know, especially in science. I think like it's hard for people to like self-promote or it's not something scientists are necessarily like naturally inclined to do. But I think having this reporting mechanism helps us like to track what's actually happening. So like if you didn't document it, you didn't do it essentially. So it is a good way to kind of look back and and you know, see your own achievements, not just not just for us, but but for yourself.

Yeah, and I think that's a great example. What we usually try to do is to apply new theoretical developments to some specific ways of doing. So we are taking it serious and trying to implement in some aspects of what we are doing. And yeah, it's really great that it's like from heavy heavy scientific quantum course goes into some actually possibly counter-intuitive but actually small steps for project to to prepare and to measure. And it gives effects, brings effects not only for project itself, but the whole nested structure of our organization and even for ecosystem as we understand it currently.

Thank you. Yeah, I blew to your point about tooting horns. I think a great role informally or possibly for an intern or for a project scale is like an internal reporter role or documentarian or agnographer role.

Someone who's just like, that was an awesome meeting. I'm going to spend a few minutes summarizing that authentically,

just by hand. Or I'm going to use LLMs to summarize that and distill that into a report. And there's so many ways in which like the people who are

still accelerating up their learning curve in a given meeting can be making those reports and asking those questions and not just be included in the culture, but actually creating it.

So that's cool.

Okay.

A closing section

towards the stack for applying active inference.

Just some quick notes and thoughts.

A lot more will unfold in the coming year, but I just wanted to footnote it in 24 and point a little bit towards some of the ways that this may be going.

During this section,

anyone, please feel free to write comments or questions in the live chat.

And then after this section, we'll have any final thoughts or comments from Alex and Blue, and then look at any questions in the live chat and address them from there.

Okay.

Whether,

when,

what,

how?

For the Institute, for active inference overall?

Whether,

whether,

to actively infer or not,

that is the question.

Whether,

tis nobler in the mind,

to suffer the slings and arrows of subject matter expertise,

or take math our arms against a sea of uncertainties,

and by expertly predicting,

bound and live with them.

When are we going to be doing that?

Well,

always,

already,

sure,

and still.

We're going to be doing it next year,

in 2025,

and we hope much beyond that too.

And,

as all,

I think,

therefore I am,

I am,
therefore I think,
type systems,
which we study,
as long as it's actively inferred to be.

So,
what really is it?
What is happening at the Institute?

We want to support,
host,
convene,
resource,
provision,
active inference,
research,
education,
and service,
forms and functions for,
modeling ecosystems of shared intelligence,
across many domains.

Those projects will always be scattered
and volatile across different developmental stages.
programs.

We will aim to stay relevantly updated with developments from outside the Institute,
about research,
companies,
tools,
and other happenings.

And,
we'll look to curate and analyze those reports of measurements into sector reports, language reports,
all different ways to make the information more accessible.

But what is it really?

Well,
beyond our core educational and open science infrastructural work,
like the ontology journal, textbook groups,
and so on,
there is a growing cadre of projects that are developing different applications of active inference.
So these are like domain or system of interest scale or user space applications for different

aspects of systems where AI has or hasn't been applied, but people are thinking about it differently with active inference.

And that includes projects and call signs like the bio firm for regenerative finance, bioregionalism,

like farm works for agriculture and IOT,

village for AI safety,

and a variety of other projects from the legal domain,

the mercuriological domain,

in the area of cognitive security,

scientific ecosystem modeling,

regional modeling,

and all these other kinds of projects that research fellows bring to the table.

So there's all these projects when people think, well, what would it really look like?

Would it be a car?

Is it a drone?

Is it precision psychiatry?

Is it resource allocation and servers?

What are those frontline applications of active inference?

And so these projects can be nurtured

on top of a robust, applicable, accessible how.

That's what people asked in the active inference symposium was, okay, applied, how?

And how is a key part of getting these projects to where they can be and where people prefer them to be.

Getting to that how and building the documentation and education, everything around it and before it, is going to continue to be a mission for us in the years to come.

In this fractal journey with the Institute as vehicle,

we see collaborators, contributors, co-explorers, and partners.

And we look towards the development of open source tools, which span the continuum from here on the left.

Let's just call it the light stack.

So this would be multimedia and interactive experiences in the no code on through low code level of technical detail,

all running in a browser or something that doesn't require installation.

So these might be variously deployed products or websites

that make graphical accessible interfaces for active inference possible.

And on the right side is the heavy stack.

So this is the sort of full-fledged enterprise grade research integrity, custom modular stack.

Starting at a place that's relevant for the project.

So just because I'm laying it out this way tentatively,

it doesn't mean that this is the entry point for every project,

but digging down, down, down to the bottom, to the basement of the bottom bunker, we have the possibility of running active inference algorithms on bare metal, silicon, and also other substrates.

Already people are working and have presented on neural substrates and physical substrates like thermocomputing.

And we can look forward to a variety of other unconventional post-Turing, post-Von Neumann architectures.

So it's important to leave the bottom floor open because we don't know the physical embodiment of what computing and cognition will look like in the future. Sure.

Working just temporarily within a computational von Neumann paradigm, we have a lot to learn from the neuroimaging community.

And I'm really thankful for Bradley Alicia and colleagues in open science for their contributions to the SAB and to the symposium, and for highlighting some of the ways in which the neuroimaging community, which is also trace it back to Carl Fristen's lab, the genesis of active inference community in certain ways, the ways that they deal with reproducibility and research integrity.

For example, having NeuroDebian, which is a variant of the Debian Linux distribution.

Maybe we could have an active Debian.

They have NeuroDesk, which is a containerization environment, flexible and scalable data analysis environment for reproducible neuroimaging.

We could have active desk, a walking desk, exercise desk, something like that, for a containerization so that open source containers and images, as well as proprietary or encrypted blobs could be provided.

And we would know that it would come to work.

Alex Arorbia in his lab has, through the NGC package, neural generative coding, has an NGC museum, which reflects all the different models that people have coded up in NGC, so that each paper can have its own repo, and then also those repos get compiled into this museum for NGC. And while we have our curated tables, Morgan highlights NeuroFedora. Thank you.

Alex Arorbia. While we have our curated tables of active inference implementations, which people can look at and see the dozens of open source repos that take time to load,

and see these repositories, some of these links might go offline, or it's hard to tell where exactly the status is, or what attributes these different models have, what kinds of inputs or outputs are they providing, and so on. And so with something like whether museum sounds a little bit static, or whether we want to have some other way to think about it, figuring out how to make generative models and ecosystem cognitive simulations reproducible is vital. I mean, it's essential for even getting

the script to run 15 minutes later on the same machine, let alone a different operating system, or a different containerization environment. So I look forward to seeing the repositories of specifications, the rendering tools like GNN, AgentMaker, GRT, P3IF, come into play, to finally, after all of that, bring it to the terminal inference algorithms, whether those are implemented in RxInfer, in PyMDP, whatever programming language, whatever way those inference algorithms are implemented,

we could have a more total understanding and customizability around going from whatever processing substrate we're working on through the operating system and containerization environment, understand what version of what generative model specification we're working off of, have reliable, if not deterministic, rendering of those generative models, and strong execution of the rendered form. This would provide the most level of customizability, and the heavy stack would underlie the graphical experiences that are hopefully enjoyed by so many on the light stack. Back to the educational side, whether leaning more on one or more on the other of these two stacks, we could have examples, guidebooks, applications, all these kinds

of interfaces available. There could be educational simulations, we could reproduce the textbook examples, other textbooks, we could reproduce papers, and support templates for people who are doing research, different kinds of simulations, math, getting into the analytical details, visualizations, games, workspaces, all these kinds of things.

At the institute level, continued development and clarification of how these core active inference functions can be built.

What are the shared cognitive computational functions that these different applications need?

That can be reflected on a shared tech tree connected to the active inference ontology.

And then understanding how that all interfaces with different applications will support collaboration.

It'll support all these different ways which people are thinking, well, how is it going to be applied?

So, they may want to jump in at the domain level or the system of interest knowing that there's something solid

and something that people could also contribute to if they wanted to underneath.

It'll be exciting to see this all unfold next year and beyond.

So, thank you all for a great tetrahedral year 2024 until the next one, which of course will be 2300.

So, Blue and Alex, what are your last thoughts or questions here? And then we'll look in the live chat.

I just wonder where the institute will be in 2300.

It will rule the world, definitely.

And I want to share my thoughts and feelings and especially after the last piece of presentation of what project in the institute currently. I have a strong feeling that engineering possibly will be the main world world for next year.

Currently, it's only in one of channels in Discord.

But possibly by the end of next year or even earlier, we will have an organizational unit of engineering because it looks like educational, research and not actually the development, but engineering should be in place

and enabled somehow by organizational structure.

Because all that genetic models, all that application, bio firms and others

definitely looks like some kind of

new entities which should burn and grow starting from

from someone's knowledge, education, from some deep research and then going into life, through engineering cycles.

And possibly currently RxInfer group, which I see is a project which has already grew

in a few beyond just RxInfer group, but a lot of other initiatives and frameworks that could be in use in place for some project specifics.

That's why engineering is a

high level concept and methods of work, which

which we also understand and putting in place in

for now project scales possibly.

I need to have more attention, more

more developments in terms of how to use engineering methods in works of all our scales.

Thank you.

That really makes me reflect on how it's obvious,

and yet, still,

it's like a blast of cold air,

that we need people with different backgrounds to be involved to contribute.

to contribute. The scientists and researchers and academics have made a lot of contributions

to active inference and also will in the time to come, and engineers may play increasingly

critical roles in supporting those core functionalities that research and application

builds and jumps off of. And there's all these other elements that we play with bringing in,

and work too to bring in, like art and cultures. And it's not for one person to see every corner of

the niche. So how we reconcile that finiteness of our own experiences and affordances and development

with the open-endedness of collective systems, and the way that we get to assemble around

the question and topic of shared intelligences while manifesting that ourselves.

Hard to put into words sometimes, slash parts that can't be put into words, but it's all

right there, and I guess everybody is just playing a different part in it.

Okay. Morgan, thank you.

Love Evolve.

Excellent. Eggnographer role. Meme lives. Egg, egg, egg.

Okay. Love Evolve wrote, please share more reflections, visions of eggnography, and especially wise

eggnography in the next phase and beyond. Okay. If either of you have a comment on that, feel free to make so. Otherwise, I'll keep that in mind.

Okay. Sharif wrote, manifest AGI. We'll wait another minute and see if anyone else writes any comments specific questions. Let's just quickly revisit what we saw while just waiting one minute for if anyone wants to write any specific questions.

[illegible]

There is a question about supporting materials in relation to the textbook group.

Okay, Morgan wrote, if we are doing a textbook reading group, do you have recommendations for supporting materials?

Do you want to give any thoughts on that?

Okay. Possibly one third about various and a lot of materials. The textbook itself contains a lot of references and the code document which enables working of this group

also has a lot of beliefs. But possibly the main thing here is the format of collective learning action.

So to be with people who also have interest to learn Active Inference and be in this group and participate in this session

and then put your own efforts to support others learning path could be much more efficient than any other like just material.

Thank you, Alex. I posted the resources link. Here, there's a table that Ali curated that provides one possible path.

But that's kind of the knowledge engineering side is resources as nodes and paths as walks on the graph.

And throughout the hundreds of questions, there's so many resources and things and somebody who is adding more fastidiously would have probably added another several factor.

So you pointed, Alex, to the group as resource that there's adding into resources, including with questions and uncertainties.

And then there's people as the resource in our path to connect.

And also, I think it's all happening so fast. So it's even it's even funny to remark on it.

But like in 2022, when we started the textbook group around this book, there was not the generative AI affordance.

Today, it's almost I see places where people are almost people, let's just say, are almost putting more synthetic content than authentic content.

And there's more people who are using LLMs to summarize a paper and then someone else uses an LLM.

You know, it's like, of course, we can go into that forever.

But in terms of how that has changed what a resource means, whereas the static resource concept was like, oh, this is an awesome stream by Ramsdead et al.

Or you should check out the paper or go to the bibliography from there or go to a scholar search and find out

what else is happening from that author or keyword.

And that kind of like library science, all of a sudden has evolved into the library being like the zoo of dreams and nightmares, where you walk in with your query.

And living artifacts come to meet you.

And so that is just pre transformed appropriately with transformers and everything like that.

And so that is just pre transformed into the landscape of learning and resources in ways that have really complex short and long term effects.

Because you're only one sentence away from, please write me an explanation of variational free energy starting from XYZ in the style of ABC.

And it's like, it's like, with that at everybody's fingertips.

Now it is trained on some of those resources and all of that.

So again, not to go too much into it, but just there are so many angles of approach that people take into active inference and so many things that people are looking to do with it.

And that would also unfold through that person's journey over the years that it's like it's and there's such a daunting amount now taken to the infinite generative level.

That it's really like, whoa, what are the right ways to learn?

But in that openness around the ways to learn and apply, that's where we have our work.

Our work.

In the uncertainty around it, not in the certainty around how to learn and apply.

Okay.

Thank you, Alex.

Thank you, Blue.

Thank you to other live chatters and viewers.

So this video should have the information that you can use.

And if not, just get in touch, whatever it is.

Okay. Bye.

Bye.

Bye.